



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Data Science Capstone Project: SpaceX Falcon 9 Launch
Success Prediction

Mudita
19 June 2026



Outline

- Executive Summary – *(Overview of project and key findings)*
- Introduction – *(Background, objectives, problem statement)*
- Methodology – *(Data collection, wrangling, EDA, SQL, Folium, Dash, models)*
- Results – *(Visualizations, SQL outputs, dashboards, predictive analysis)*
- Conclusion – *(Insights, implications, future work)*
- Appendix – *(Extra charts, queries, code snippets)*

Executive Summary

SpaceX has revolutionized the space industry by reducing launch costs through reusable Falcon 9 boosters. This project analyzes historical Falcon 9 launch data to identify factors influencing booster landing success and uses machine learning to predict first-stage recovery outcomes, helping estimate launch costs and reusability.

- **Summary of Methodologies:**

- **Data Collection:** Gathered launch data using the SpaceX API and Wikipedia web scraping.
- **Data Wrangling:** Cleaned, transformed, and prepared datasets for analysis.
- **Exploratory Data Analysis (EDA):** Performed statistical analysis using SQL queries and visualizations.
- **Interactive Analytics:** Developed Folium maps and Plotly Dash dashboards to explore launch patterns and outcomes.
- **Predictive Modeling:** Built and evaluated Logistic Regression, SVM, Decision Tree, and KNN models to predict Falcon 9 first-stage landing success.

- **Summary of Results:**

- **Dataset Preparation:** Created a structured and analysis-ready Falcon 9 launch dataset.
- **EDA Findings:** Identified key trends related to payload mass, launch sites, booster versions, and mission outcomes.
- **SQL Analysis:** Validated launch site distributions, payload statistics, and landing outcome patterns.
- **Geospatial Analysis:** Used Folium maps to visualize launch site locations and their geographical significance.
- **Interactive Analytics:** Developed Plotly Dash dashboards for dynamic exploration of launch success rates and mission characteristics.
- **Machine Learning Outcome:** Logistic Regression achieved the highest test accuracy (**83.3%**) and emerged as the best model for predicting first-stage landing success.

Introduction

1. Project background and context:

- SpaceX has significantly reduced launch costs by **reusing Falcon 9 first-stage boosters**, lowering launch costs to approximately **\$62M compared to competitors' ~\$165M**.
- The successful recovery and reuse of the first stage is a key factor influencing **launch cost, operational efficiency, and mission sustainability**.
- As a Data Scientist working for the fictional company **Space Y**, the objective was to analyze historical Falcon 9 launch data collected from the **SpaceX API and Wikipedia**.
- Using data analytics and machine learning, the project aims to identify factors affecting booster landing success and predict whether the first stage will be successfully recovered and reused.

2. Problems you want to find answers:

- Which launch sites contribute the most to successful launches and landings?
- How do payload mass, orbit type, and booster version influence landing success?
- What patterns and trends can be identified from historical Falcon 9 missions?
- Can machine learning models accurately predict first-stage landing outcomes and support launch cost estimation?

Section 1

Methodology

Methodology

- **Data Collection:**

- ✓ Collected Falcon 9 launch data from the **SpaceX API** and **Wikipedia web scraping**, and integrated the data into a unified dataset.

- **Data Wrangling:**

- ✓ Cleaned and preprocessed the dataset by handling missing values, standardizing formats, and creating relevant features for analysis.

- **Exploratory Data Analysis (EDA):**

- ✓ Performed exploratory analysis using **SQL queries** and **visualizations** to identify patterns and relationships among launch variables.

- **Interactive Visual Analytics:**

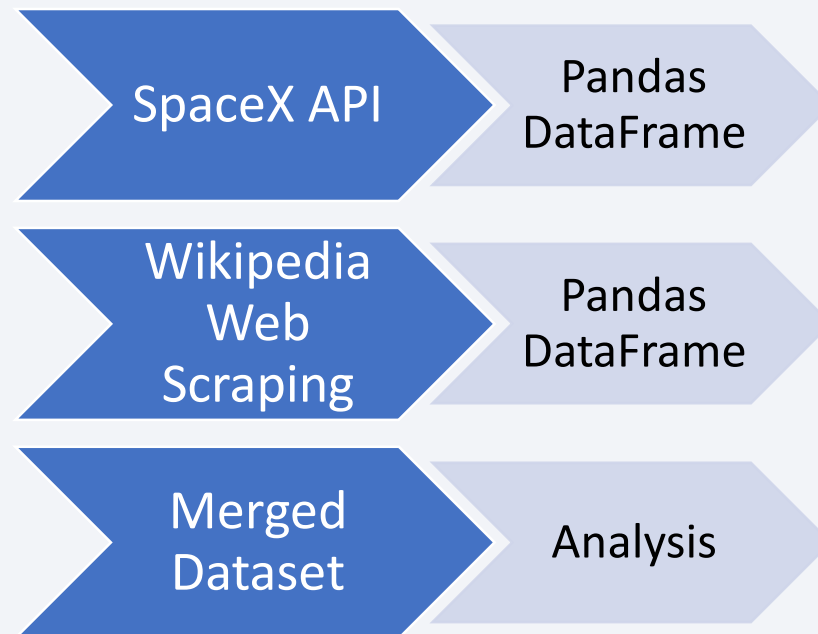
- ✓ Developed **Folium maps** and **Plotly Dash dashboards** to explore launch site locations, mission outcomes, and launch performance interactively.

- **Predictive Analysis:**

- ✓ Built classification models including **Logistic Regression**, **SVM**, **Decision Tree**, and **KNN** to predict first-stage landing success.
- ✓ Applied **GridSearchCV** for hyperparameter tuning and evaluated model performance using **accuracy scores** and **confusion matrices**.

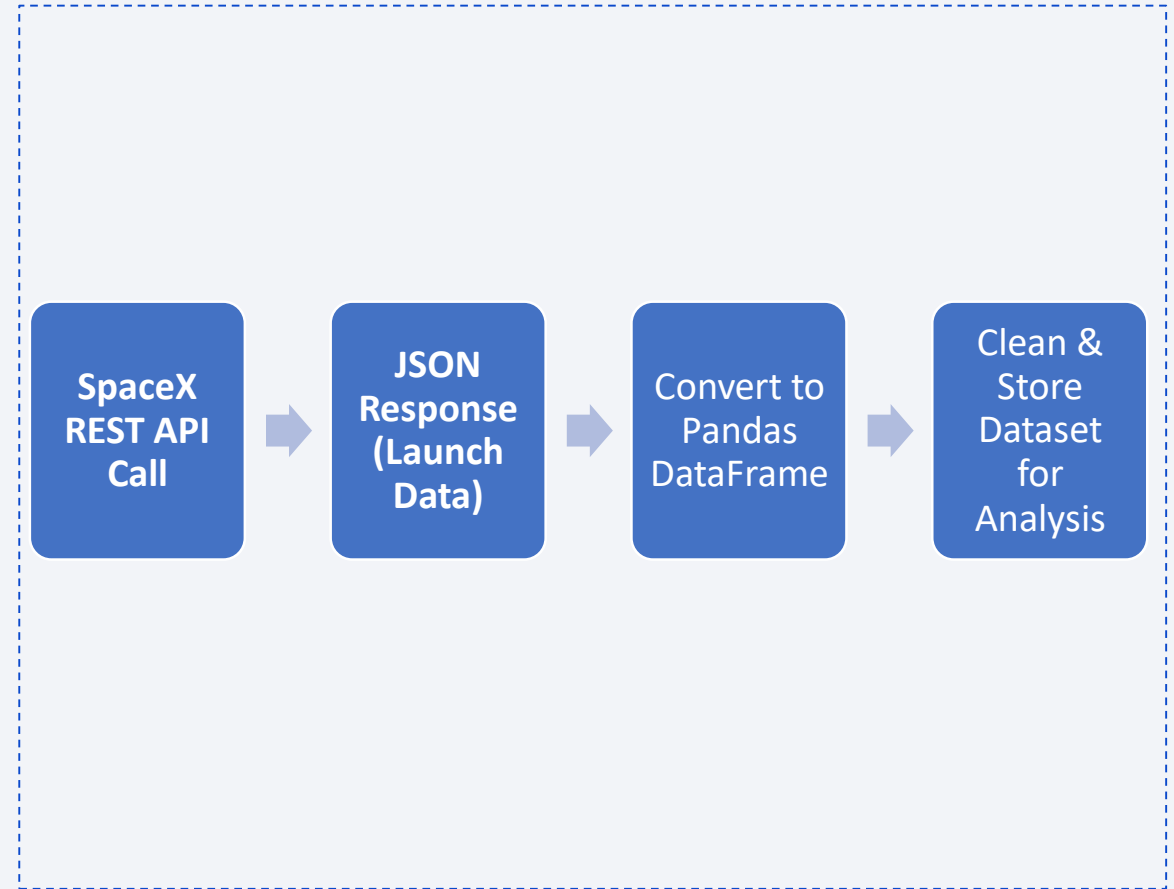
Data Collection

- How data sets were collected.
 - ✓ Launch data retrieved via the official **SpaceX API** (JSON format).
 - ✓ Additional launch records scraped from **Wikipedia** tables using BeautifulSoup.
 - ✓ Both datasets converted into **Pandas DataFrames** and merged into a single dataset for analysis.



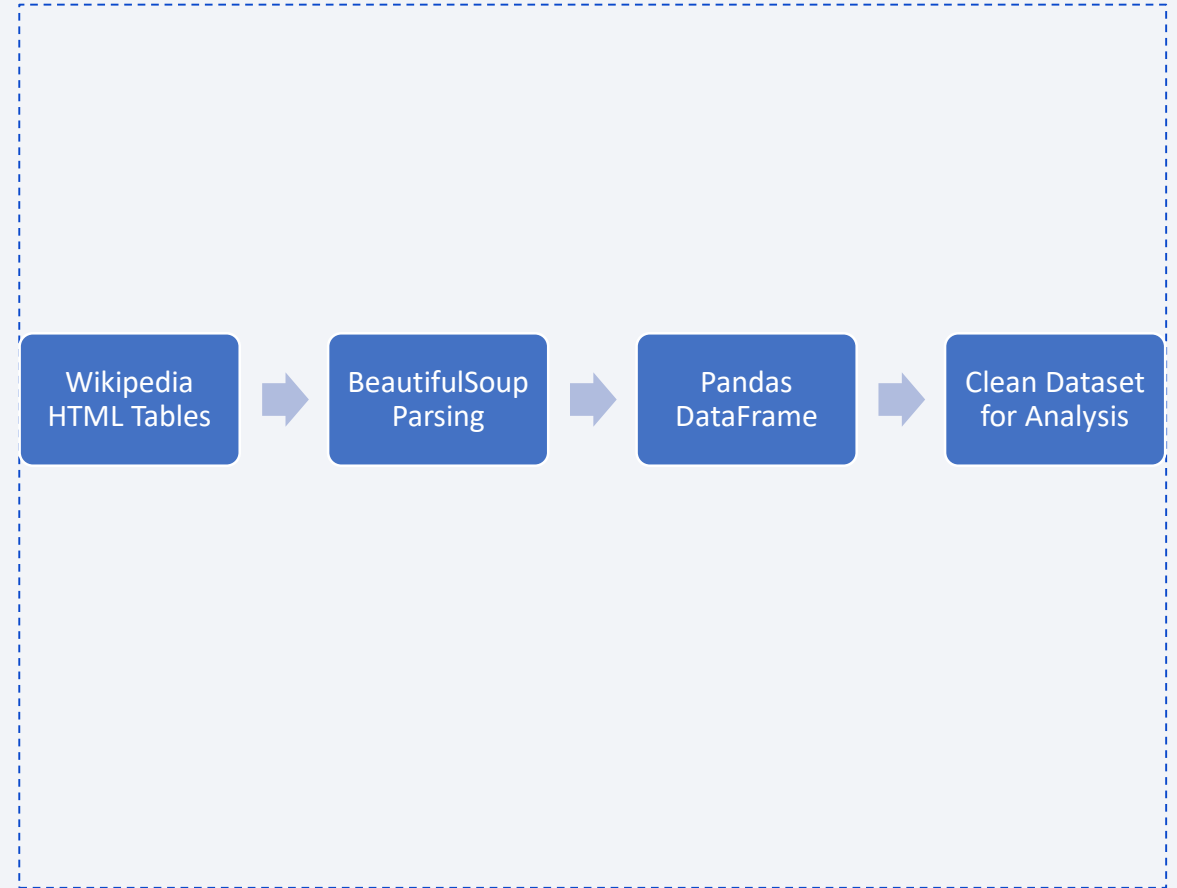
Data Collection – SpaceX API

- **Launch data** retrieved via **SpaceX REST API** (JSON format).
- Extracted attributes such as *Flight Number*, *Launch Site*, *Payload Mass*, *Orbit*, *Outcome*.
- Converted JSON responses into **Pandas DataFrames** for analysis.
- **GitHub URL** of the completed SpaceX API calls **notebook(code + output)**:
<https://github.com/Mudita13/SpaceX-API-data-collection/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



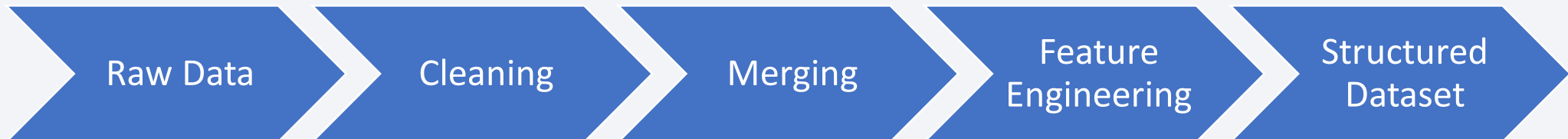
Data Collection - Scraping

- Scraped launch records from **Wikipedia HTML tables** using **BeautifulSoup**.
- Extracted attributes such as *Payload Mass, Orbit, Launch Site, Outcome*.
- Converted scraped tables into **Pandas DataFrames** for integration with API data.
- **GitHub URL** of the completed web scraping **notebook(code + output)**:
<https://github.com/Mudita13/SpaceX-data-collection--web-scraping/blob/main/jupyter-labs-webscraping.ipynb>



Data Wrangling

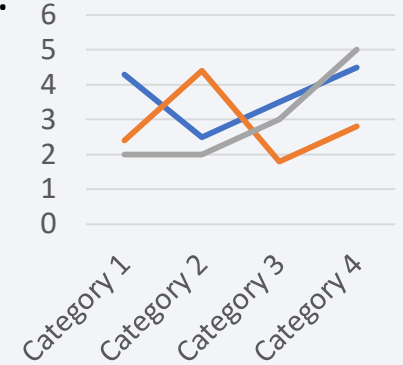
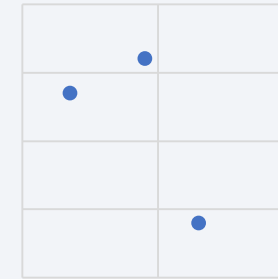
- How data were processed :
 - **Cleaned missing values** and **standardized** column formats.
 - **Merged** SpaceX API and Wikipedia scraped **datasets** into one **DataFrame**.
 - **Engineered new features** (*Booster Version Category, Landing Outcome*).
 - **Structured final tables** for **SQL queries** and **visualizations**.
- **GitHub URL** of completed data wrangling **notebook** : <https://github.com/Mudita13/SpaceX-Data-wrangling/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>



EDA with Data Visualization

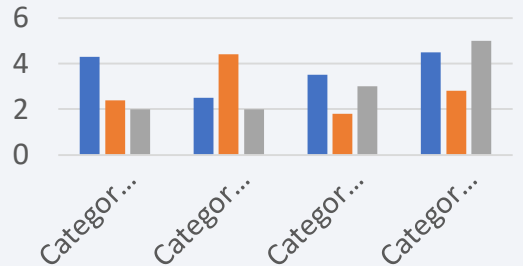
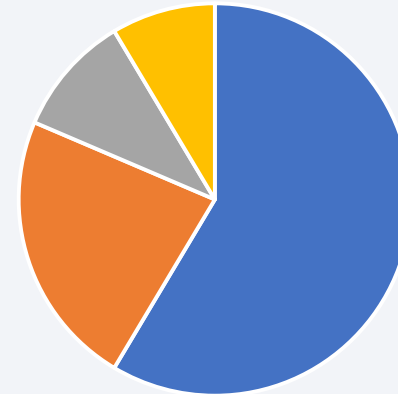
- Summary of charts plotted:

- **Scatter plots** → To analyze correlation between payload mass and landing success.
- **Bar charts** → To compare launch outcomes across different sites.
- **Pie charts** → To show distribution of orbit types.
- **Line charts** → To observe trends in launches over time.



- Why these charts:

- **Scatter plots** reveal **relationships** between variables.
- **Bar charts** highlight **categorical comparisons**.
- **Pie charts** show **proportions** clearly.
- **Line charts** capture **temporal trends**.



- **GitHub URL** of your completed **EDA with data visualization notebook** :
[https://github.com/Mudita13/SpaceX-EDA-with-Visualization/blob/main/edadataviz%20\(1\).ipynb](https://github.com/Mudita13/SpaceX-EDA-with-Visualization/blob/main/edadataviz%20(1).ipynb)

EDA with SQL

- Summary the SQL queries performed:
 - **DISTINCT** → Listed unique launch sites.
 - **LIKE with LIMIT** → Displayed 5 records where launch sites begin with “CCA”.
 - **SUM** → Calculated total payload mass for NASA (CRS).
 - **AVG** → Found average payload mass for booster version *F9 v1.1*.
 - **MIN** → Identified first successful ground pad landing date (2015-12-22).
 - **DISTINCT with conditions** → Listed boosters with successful drone ship landings and payload mass between 4000–6000 kg.
 - **GROUP BY with COUNT** → Counted successful vs. failed mission outcomes.
 - **Subquery with MAX** → Found boosters carrying maximum payload mass.
 - **SUBSTR (Date)** → Extracted months in 2015 with failed drone ship landings.
 - **BETWEEN with ORDER BY** → Ranked landing outcomes between 2010-06-04 and 2017-03-20.
- **GitHub URL** of completed **EDA with SQL notebook** : https://github.com/Mudita13/SpaceX-EDA-with-SQL/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Map objects created:
 - ✓ **Markers** → Placed at each launch site to show exact location.
 - ✓ **Circles** → Represented payload mass visually (size proportional to payload).
 - ✓ **Popups** → Displayed launch site names and outcomes when clicked.
 - ✓ **Lines/Polylines** → Connected launch sites to orbits for trajectory visualization.
- Why these objects were added?
 - Markers make locations easy to identify.
 - Circles provide quick visual comparison of payload sizes.
 - Popups add interactivity and context for each site.
 - Lines illustrate relationships between launch sites and orbital paths.
- **GitHub URL of completed interactive map with Folium map notebook:**
<https://github.com/Mudita13/SpaceX-Interactive-Visual-Analytics-with-Folium>

Build a Dashboard with Plotly Dash

- **Summary of Plots/Graphs and Interactions Added :**
 - ✓ **Dropdown menu** → Select launch site dynamically.
 - ✓ **Range slider** → Adjust payload mass range interactively.
 - ✓ **Pie chart** → Visualize success counts by site or success vs. failure for a chosen site.
 - ✓ **Scatter plot** → Show correlation between payload mass and launch outcome, color-coded by booster version.
- **Why These Were Added?**
 - Dropdown enables flexible site-specific analysis.
 - Range slider allows exploration of payload ranges and their effect on outcomes.
 - Pie chart provides clear proportions of successful launches.
 - Scatter plot reveals correlations between payload and success, with booster comparison.
- **GitHub URL** of completed **Plotly Dash lab** : <https://github.com/Mudita13/SpaceX-Dashboard-with-Plotly>

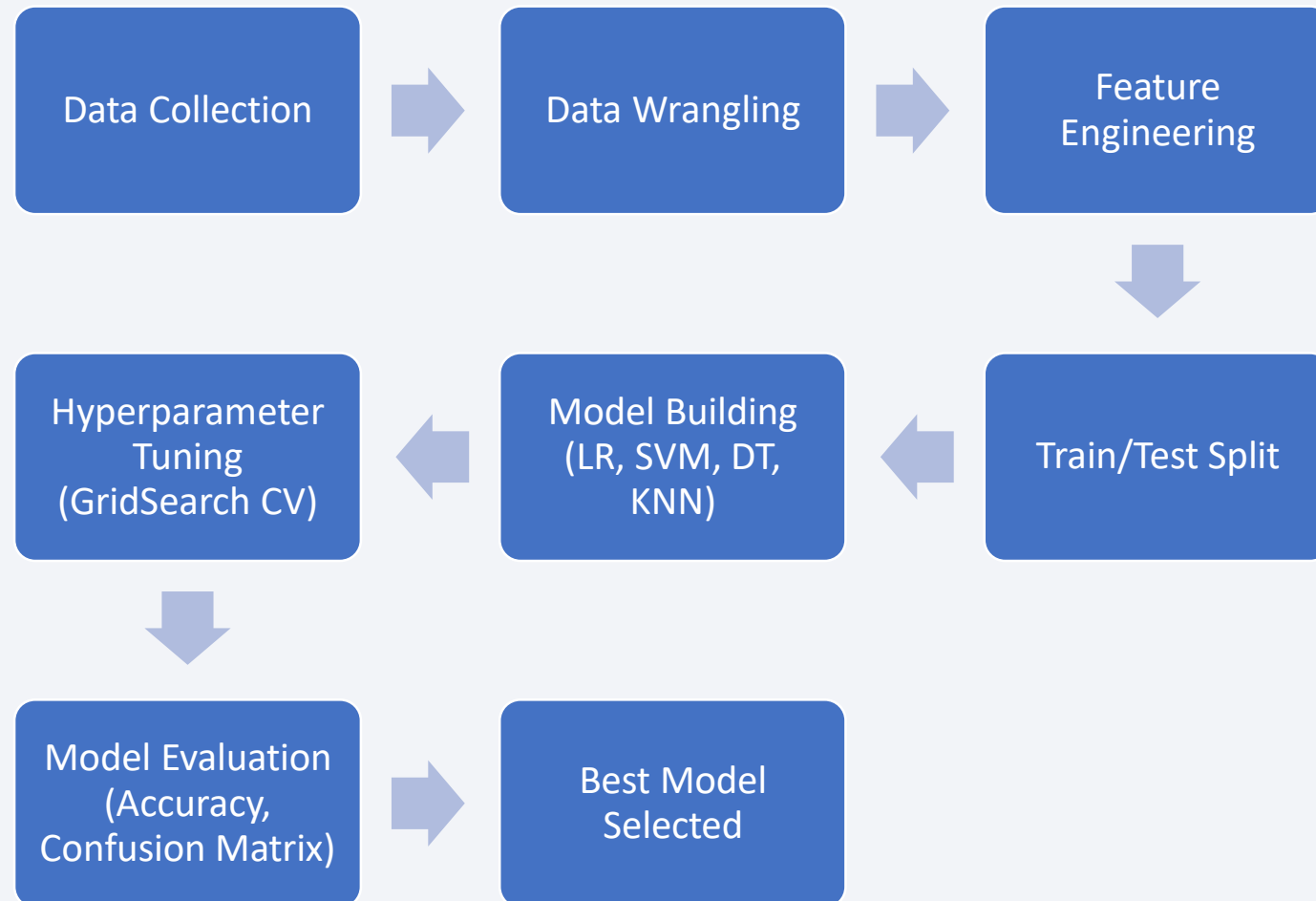
Predictive Analysis (Classification)

Model Development Process:

- **Data Preparation:** Cleaned and standardized features for model input.
 - **Model Building:** Applied **Logistic Regression, SVM, Decision Tree, and KNN** to predict Falcon 9 first-stage landing success..
 - **Hyperparameter Tuning:** Used **GridSearchCV (cv=10)** to optimize parameters.
 - **Evaluation:** Compared models using **cross-validation accuracy, test accuracy, and confusion matrices**.
 - **Improvement:** Iterated with tuned parameters to enhance performance.
 - **Best Model:** **Logistic Regression** identified as the most stable and interpretable, achieving **≈ 83% test accuracy**.
-
- **GitHub URL** of completed **predictive analysis lab**: [https://github.com/Mudita13/SpaceX-ML-Prediction-Classification/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20\(1\).ipynb](https://github.com/Mudita13/SpaceX-ML-Prediction-Classification/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb)

Predictive Analysis (Classification) - Flowchart

Model Development Flowchart:



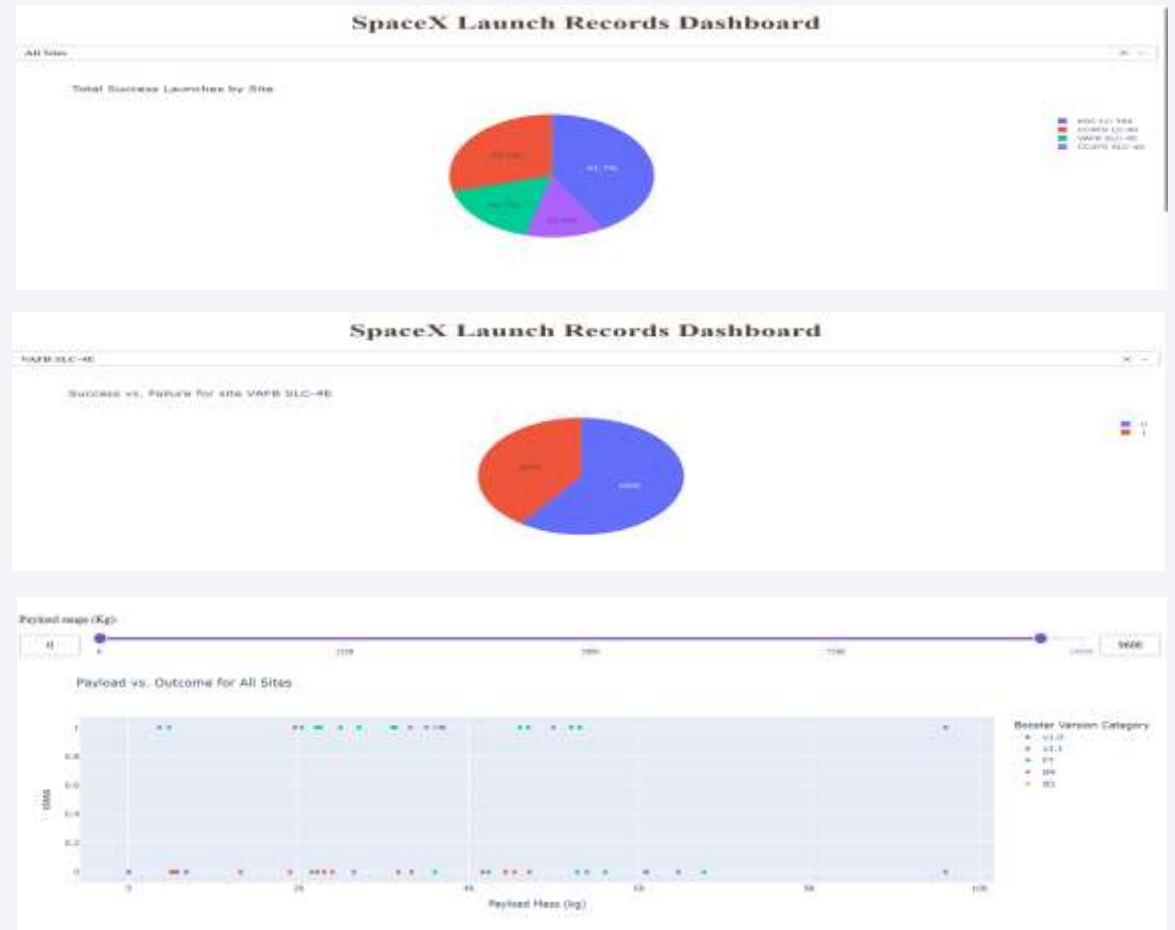
Results - (Exploratory Data Analysis & Interactive Analytics)

- **Exploratory Data Analysis (EDA) Results**

- ✓ Payload distribution patterns across missions
- ✓ Launch success rates across sites
- ✓ Orbit frequency analysis
- ✓ Site-specific performance comparisons

- **Interactive Analytics Features**

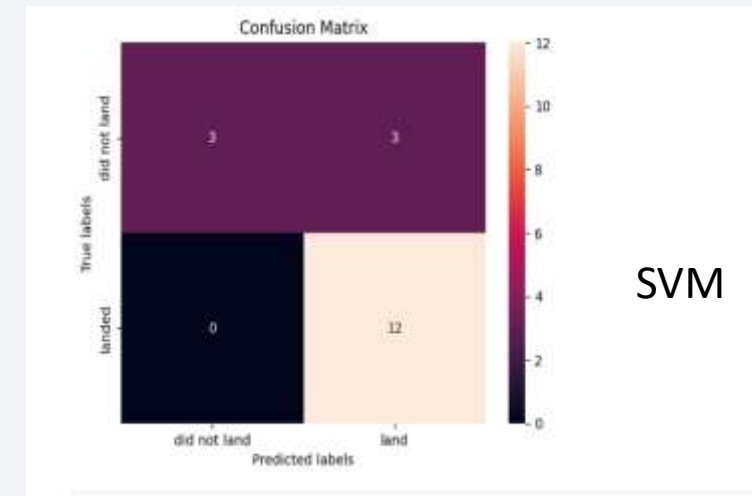
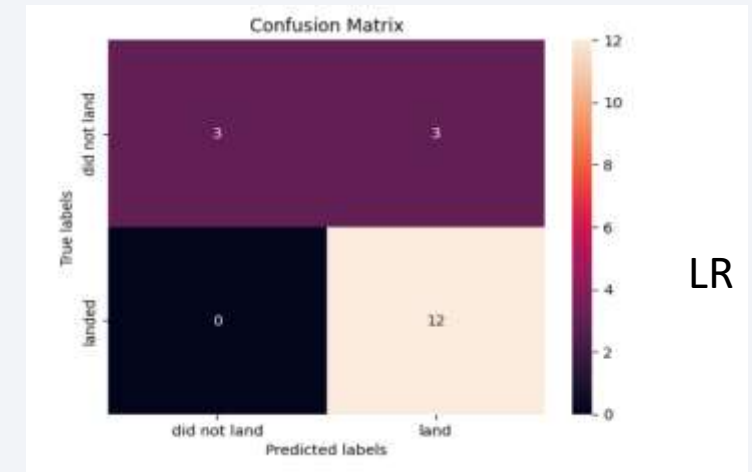
- ✓ Dropdown for site selection
- ✓ Range slider for payload filtering
- ✓ Pie chart for success counts
- ✓ Scatter plot for payload vs. outcome with booster categories

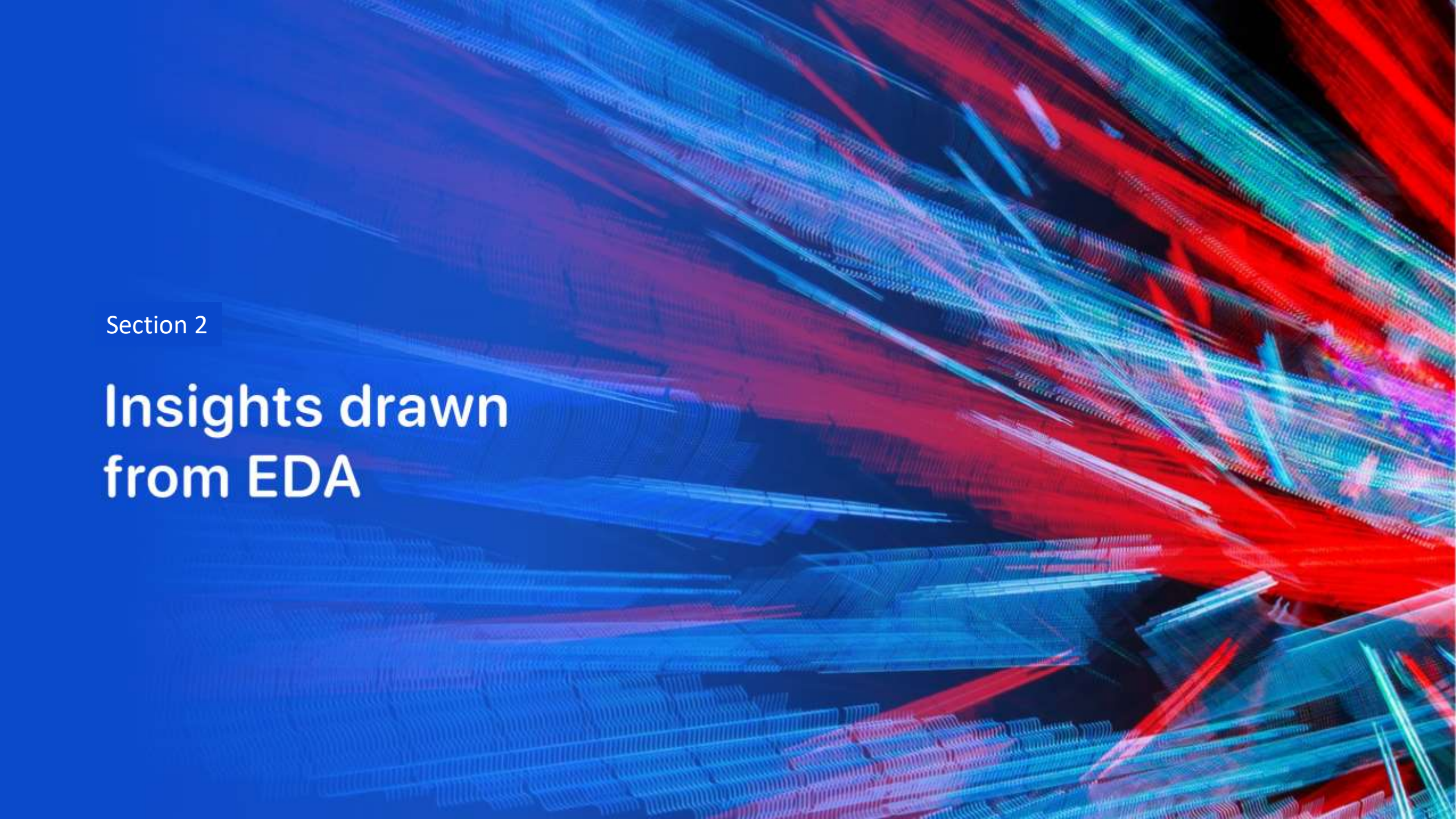


Results - (Predictive Analysis)

- **Predictive Analysis Results:**

- ✓ Built and evaluated classification models: **Logistic Regression, SVM, Decision Tree, and KNN.**
- ✓ Applied **GridSearchCV (cv = 10)** for hyperparameter tuning to optimize performance.
- ✓ Models were assessed using **cross-validation accuracy, test accuracy, and confusion matrices** to compare outcomes.
- ✓ **All models achieved ~83.3% test accuracy** with identical confusion matrices.
- ✓ **Logistic Regression selected as best method** due to stability, simplicity, and interpretability for predicting Falcon 9 first stage landing success.



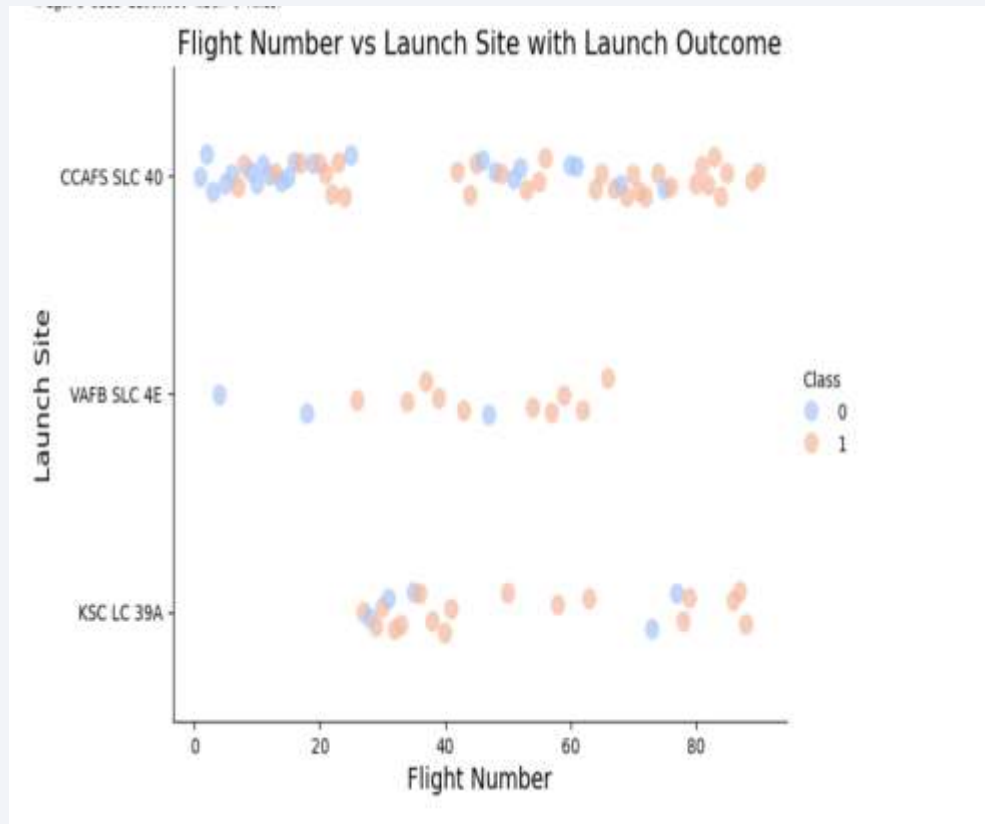
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue, red, and cyan on the right. Overlaid on these streaks is a faint, semi-transparent grid of small squares, creating a complex, layered visual effect.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

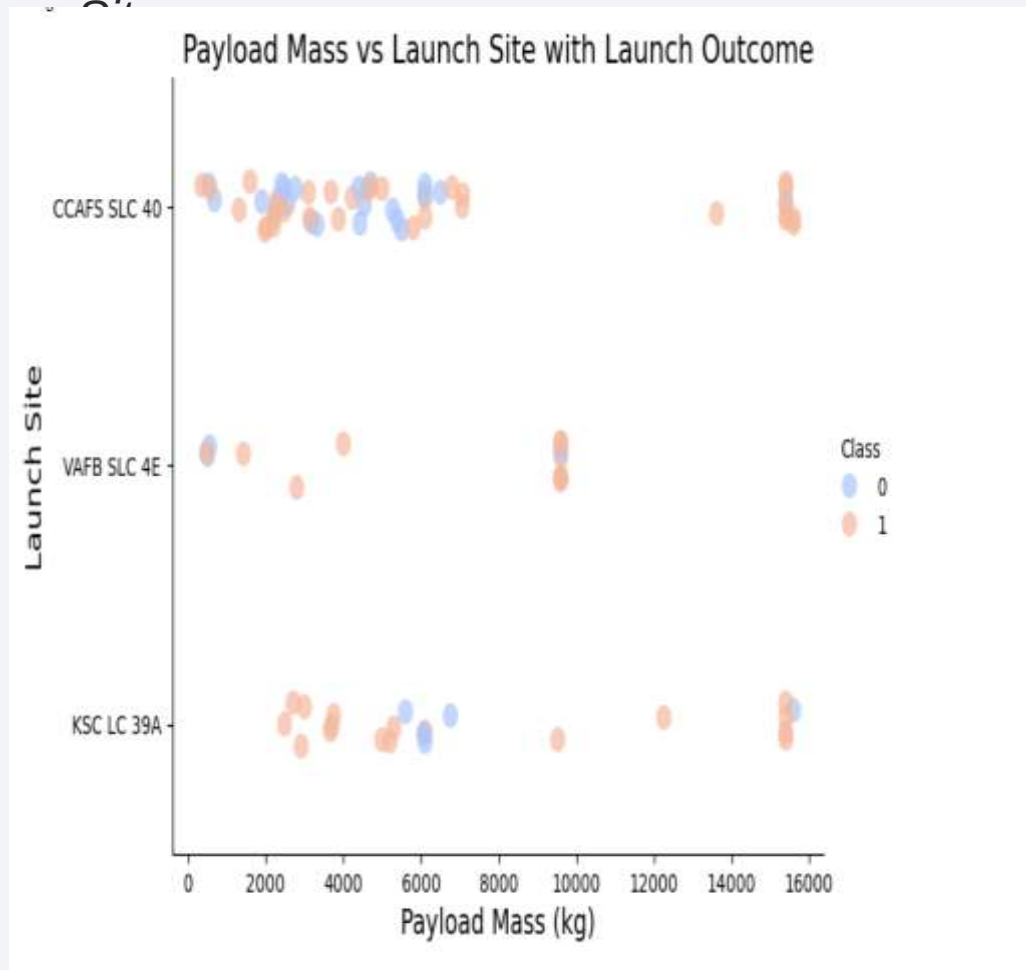
- **Scatter plot of *Flight Number* vs. *Launch Site*:**



- **Explanation :**
 - ✓ **Flight Number (X-axis):** Represents the chronological order of Falcon 9 launches.
 - ✓ **Launch Site (Y-axis):** Shows three sites — CCAFS SLC-40, VAFB SLC-4E, and KSC LC-39A.
 - ✓ **Color coding (Outcome):** Orange = Success, Blue =Failure.
- **Insights from the plot:**
 - **CCAFS SLC-40:** Early launches had mixed outcomes, with both successes and failures scattered across flight numbers. Later launches show improved reliability, reflecting SpaceX's learning curve.
 - **KSC LC-39A:** Launches occur at higher flight numbers, when SpaceX's technology was more mature. Strong concentration of successful landings, indicating this site became highly reliable.
 - **VAFB SLC-4E:** Fewer launches overall. Outcomes are mixed, but limited data makes trends less clear.
 - **Overall Trend:** As flight numbers increased, the proportion of successful landings grew. This demonstrates SpaceX's **iterative improvements and operational maturity** over time.

Payload vs. Launch Site

- **Scatter plot of *Payload Mass vs. Launch***



- **Explanation:**

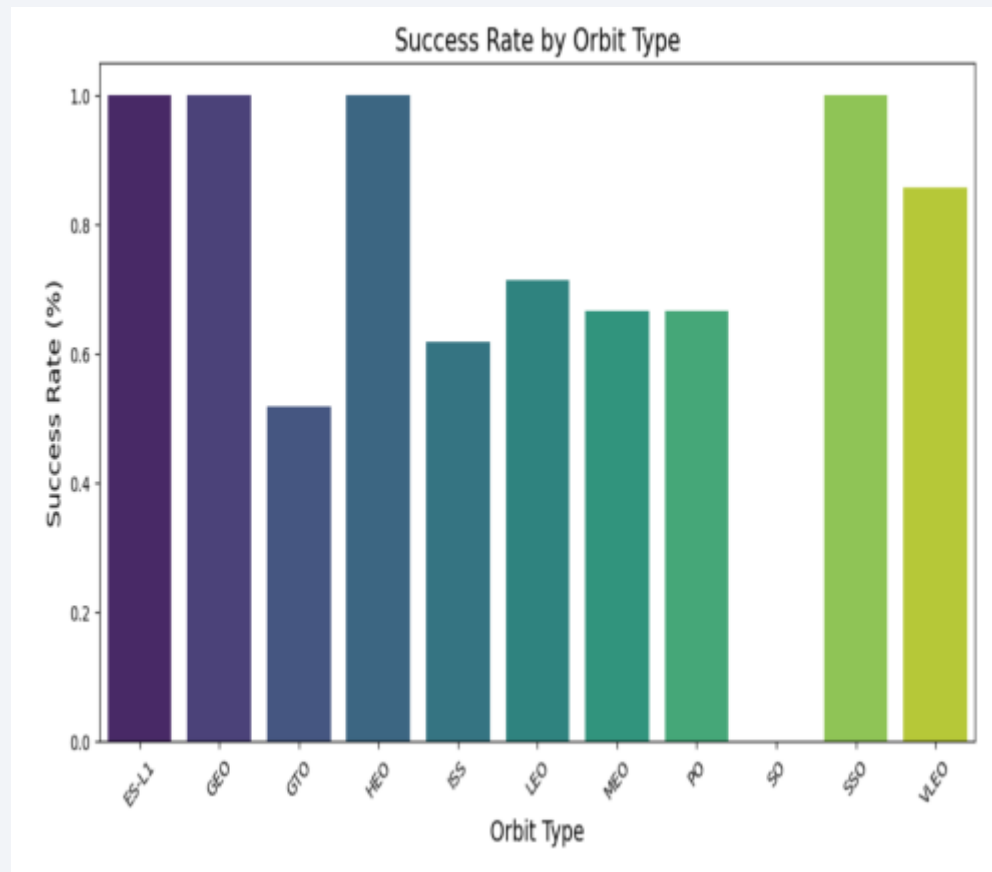
- ✓ **Payload (X-axis):** Represents the mass of the payload carried by Falcon 9 launches.
- ✓ **Launch Site (Y-axis):** Shows three sites — **CCAFS SLC-40**, **VAFB SLC-4E**, and **KSC LC-39A**.
- ✓ **Color coding (Outcome):** Each point indicates whether the mission was a success or failure, with colors distinguishing the outcome.

- **Insights from the plot:**

- **CCAFS SLC-40:** Handles a wide range of payloads. Early launches show mixed outcomes, but as payloads increased, reliability improved — reflecting SpaceX's growing confidence in this site for diverse missions.
- **KSC LC-39A:** Launches here are concentrated at higher payload values. The outcomes are strongly successful, showing that this site became the preferred choice for heavy-lift missions once Falcon 9 technology matured.
- **VAFB SLC-4E:** Fewer launches overall, mostly in the medium payload range. Outcomes are mixed, and limited data makes it harder to establish a clear trend compared to the other sites.
- **Overall Trend:** As payloads increased, the proportion of successful missions grew. This demonstrates SpaceX's iterative improvements and operational maturity, with heavier payloads increasingly launched from the most reliable site (KSC LC-39A).

Success Rate vs. Orbit Type

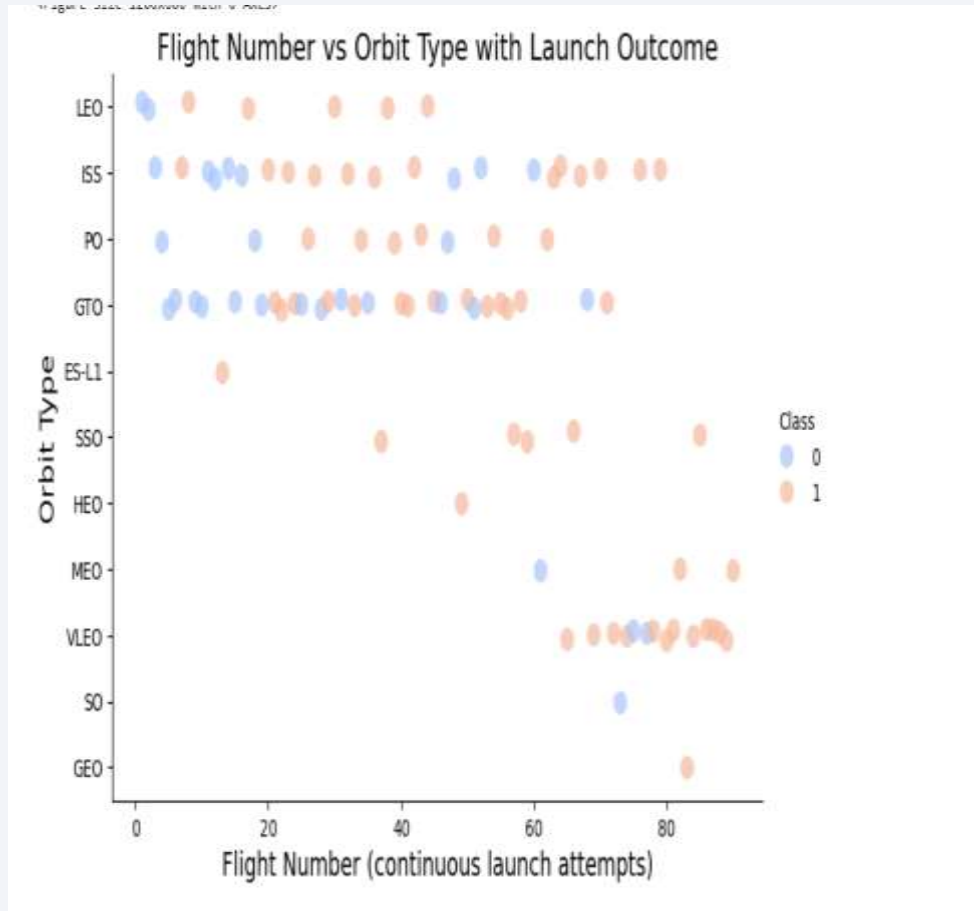
- **Bar chart** for the *Success Rate of each Orbit Type*:



- **Explanation:**
 - ✓ **X-axis:** Orbit types (11 categories).
 - ✓ **Y-axis:** Average success rate (calculated as mean of Class).
 - ✓ **Bars:** Show how reliable Falcon 9 launches were for each orbit type.
- **Insights:**
 - **Near-Perfect Success (~100%)**
 - **ES-L1, GEO, HEO, SSO:** These orbit types remain the most reliable, showing almost flawless mission success. These are mature, well-understood orbits with standardized mission designs.
 - **High Success but Not Perfect (70–90%)**
 - **LEO (~70%):** Most common orbit, reliable but exposed to risks due to high mission volume.
 - **ISS (~80–85%):** Human-rated missions prioritize safety, keeping success high.
 - **VLEO (~80%):** Newer orbit type, showing strong early performance.
 - **Moderate Success (~65%)**
 - **MEO (~65%), PO (~65%):** These orbits are less frequently used. The moderate success rate suggests technical challenges or fewer missions to stabilize reliability.
 - **Lowest Success (0–50%)**
 - **SO (0%):** No successful missions recorded in this dataset.
 - **GTO (~50%):** Transfer orbit complexity makes this the riskiest category. Failures often occur during upper-stage burns or orbital insertion.

Flight Number vs. Orbit Type

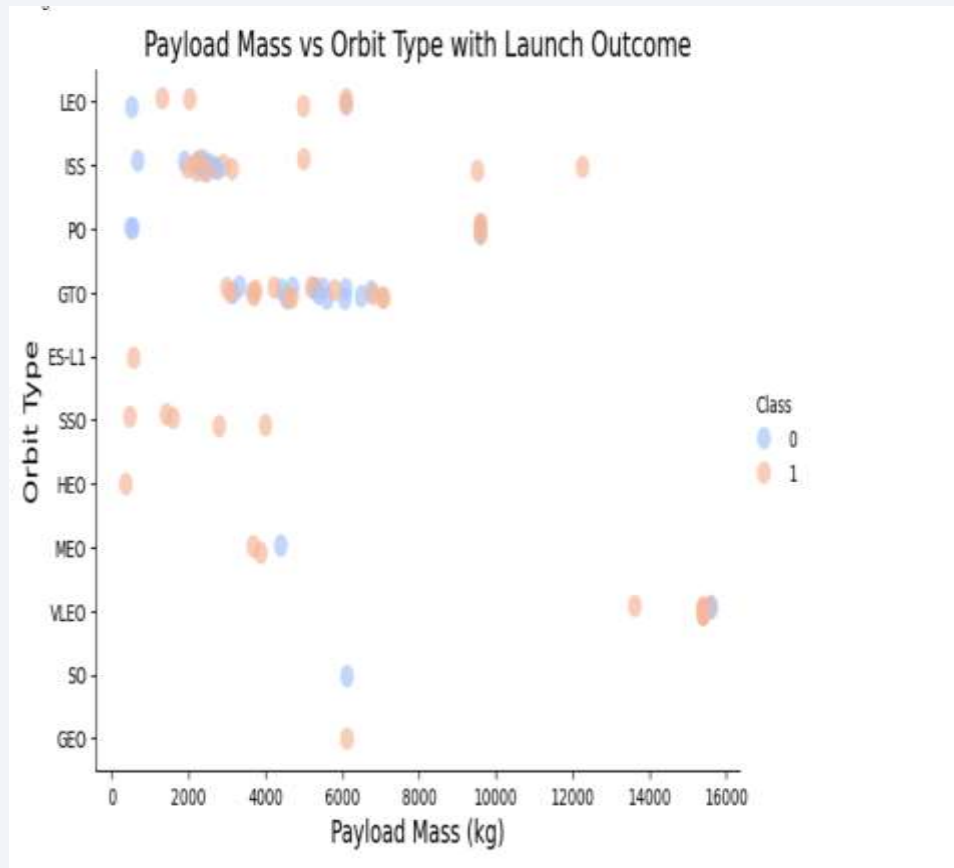
- Scatter plot of *Flight number vs. Orbit type*:



- **Explanation:**
 - ✓ **X-axis:** Flight Number (sequential launch attempts).
 - ✓ **Y-axis:** Orbit Type (LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, GEO).
 - ✓ **Color (Class):** Launch outcome — one color for success, another for failure.
- **Insights:**
 - **LEO:** Early flights mixed, later flights mostly successful → clear learning curve.
 - **ISS:** Consistently high success → human-rated missions prioritize safety.
 - **SSO:** Stable success across flights → standardized mission design.
 - **GTO:** Success/failure scattered → orbit complexity limits improvement.
 - **MEO/PO:** Moderate reliability (~65%) → fewer missions, less maturity.
 - **VLEO (~80%):** Success achieved only after flight number 60 → reflects maturity and improved reliability in later missions.
 - **SO:** No successes → high risk/experimental.
 - **(ES-L1, HEO, GEO):** Limited data but strong success when attempted.

Payload vs. Orbit Type

- Scatter plot of *Payload vs. Orbit Type*:



- **Explanation:**

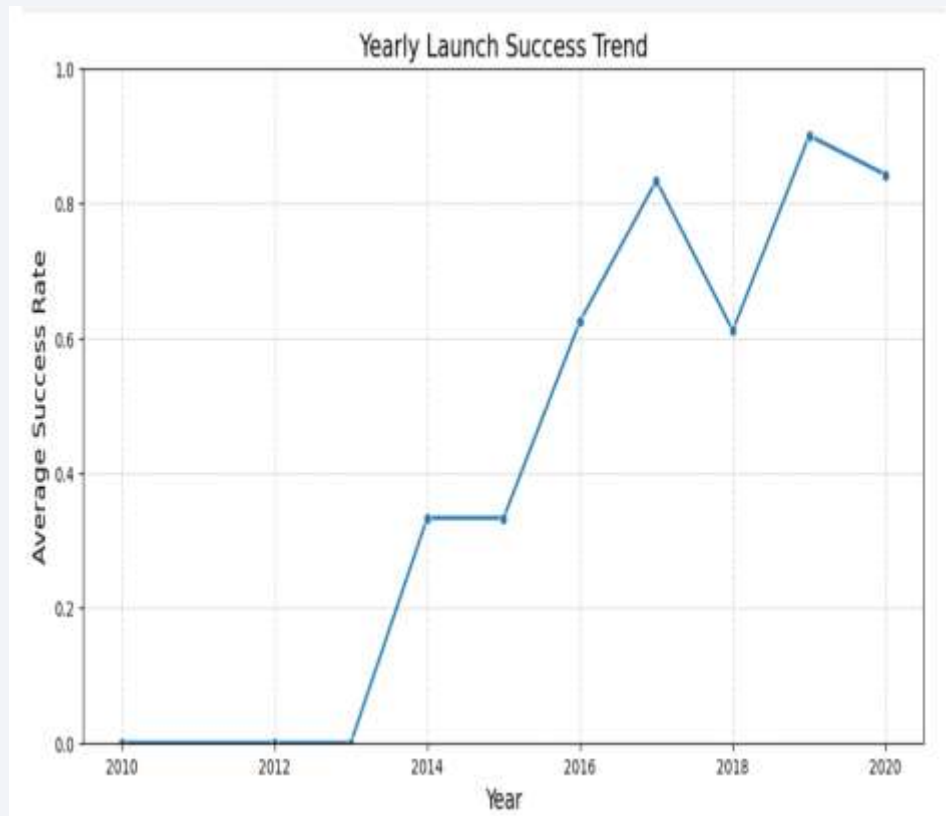
- ✓ **X-axis:** Payload Mass (kg), ranging from 0 to ~16,000.
- ✓ **Y-axis:** Orbit Type (LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, GEO).
- ✓ **Color:** Launch outcome (Success vs Failure).

- **Insights:**

- **Low-Mass Orbits (ISS, PO, SSO, HEO, ES-L1):** Primarily used for scientific, Earth observation, and station-resupply missions; these launches generally carry payloads below 4,000 kg and exhibit high success rates.
- **Medium-Mass Orbits (LEO, MEO, GTO, GEO):** Payloads typically range from 3,000–7,000 kg, with **GTO dominating the dataset**, reflecting the high demand for commercial satellite deployments.
- **High-Mass Orbits (VLEO):** Carry the heaviest payloads (14,000–16,000 kg) and show predominantly successful launch outcomes, highlighting strong heavy-lift launch capabilities.
- **Launch Outcome Trend:** Successful and failed launches occur across similar payload mass ranges, indicating that **orbit type and mission profile are more influential than payload mass alone** in determining launch success.

Launch Success Yearly Trend

- **Line Chart** of *Yearly Average Success Rate*:



- **Explanation of the Line Chart:**

- ✓ **X-axis:** Years (2010–2020).
- ✓ **Y-axis:** Average Success Rate (0.0–1.0).
- ✓ **Trend:** The line shows how launch success rates evolved year by year.

- **Insights:**

- **2010–2013:** Success rate at **0.0** → early years marked by failures or no successful launches.
- **2014–2015:** Improvement to **~0.33** → first consistent successes begin.
- **2016:** Sharp rise to **~0.63** → technology and processes maturing.
- **2017:** Peak near **0.83** → strong reliability achieved.
- **2018:** Dip to **~0.61** → setbacks highlight ongoing challenges.
- **2019:** Recovery to **~0.9** → highest success rate, showing operational maturity.
- **2020:** Slight decline to **~0.85** → still very high, proving sustained reliability.

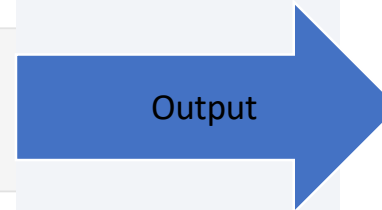
All Launch Site Names

- Find the names of the unique launch sites:

```
Display the names of the unique launch sites in the space mission

%%sql
SELECT DISTINCT "Launch_Site"
FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.
```



Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Explanation:**
 - ✓ The query uses **SELECT DISTINCT** to retrieve only **unique launch site names** from the SPACEXTABLE.
 - ✓ **DISTINCT** removes duplicate entries, ensuring each launch site appears only once.
 - ✓ This helps identify all launch locations used across SpaceX missions and provides a clean summary of launch infrastructure.

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA':

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT *  
FROM SPACEXTABLE  
WHERE "Launch_Site" LIKE 'CCA%'  
LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Explanation:**
 - ✓ The query filters records where the **Launch_Site** starts with "CCA" using the LIKE 'CCA%' operator.
 - ✓ % is a wildcard that matches any characters after CCA.
 - ✓ LIMIT 5 restricts the output to the first five matching records.

Total Payload Mass

- The total payload carried by boosters from NASA:

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
SELECT SUM("PAYLOAD_MASS__KG_") AS TotalPayloadMass
FROM SPACEXTABLE
WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

TotalPayloadMass
45596

- Explanation:
 - ✓ SUM("PAYLOAD_MASS__KG_") adds together the payload masses (in kilograms) from all matching missions.
 - ✓ The WHERE clause filters the dataset to include only records where the customer is **NASA (CRS)**.
 - ✓ Result is Total Payload Mass = 45,596 kg

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG("PAYLOAD_MASS__KG_") AS AvgPayloadMass
FROM SPACEXTABLE
WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

AvgPayloadMass

2928.4

- **Explanation**
 - ✓ AVG("PAYLOAD_MASS__KG_") computes the mean payload mass (in kilograms) across all matching launches.
 - ✓ The WHERE clause filters the data to include only missions flown using **F9 v1.1**.
 - ✓ **Result is Average Payload Mass = 2,928.4 kg**
 - ✓ **Key Insight** - On average, **Falcon 9 v1.1** missions carried approximately **2.93 metric tons** of payload.

First Successful Ground Landing Date

- The dates of the first successful landing outcome on ground pad:

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%%sql
SELECT MIN("Date") AS FirstGroundPadLandingDate
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
Done.
```

FirstGroundPadLandingDate
2015-12-22

- **Explanation**
 - ✓ The query identifies the **earliest date** on which a booster achieved a **successful ground pad landing**.
 - ✓ MIN("Date") returns the earliest (minimum) date from the filtered records.
 - ✓ The WHERE clause selects only missions where the Landing_Outcome was **"Success (ground pad)"**.
 - ✓ **Result is First Successful Ground Pad Landing Date: 2015-12-22**
 - ✓ **Key Insight - 22 December 2015** marks a major milestone in SpaceX history—the first successful Falcon 9 booster landing on a ground landing pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

```
List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

%%sql
SELECT DISTINCT "Booster_Version"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (drone ship)'
  AND "PAYLOAD_MASS_KG_" > 4000
  AND "PAYLOAD_MASS_KG_" < 6000;

* sqlite:///my_data1.db
Done.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2
```

- Explanation:**

- ✓ The query retrieves the **unique booster versions** that meet two conditions:
 1. Achieved a **successful drone ship landing** (Landing_Outcome = 'Success (drone ship)').
 2. Carried a **payload mass between 4,000 kg and 6,000 kg**.
- ✓ DISTINCT ensures that each booster version appears only once in the result.
- ✓ The AND operator combines the landing success and payload mass criteria.

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes:

List the total number of successful and failure mission outcomes

```
3]: %%sql
SELECT "Mission_Outcome", COUNT(*) AS OutcomeCount
FROM SPACEXTABLE
GROUP BY "Mission_Outcome";

* sqlite:///my_data1.db
Done.
```

Output

Mission_Outcome	OutcomeCount
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Explanation:**
 - ✓ The query counts the number of missions for each **Mission_Outcome** category.
 - ✓ COUNT(*) calculates how many records belong to each outcome type.
 - ✓ GROUP BY "Mission_Outcome" groups missions based on their outcome before counting them.
- Insight:**
 - ✓ **Mission success rate is exceptionally high**, with **99+ successful missions** compared to only **1 in-flight failure**.
 - ✓ The presence of two "Success" categories likely indicates a **data quality issue** (e.g., extra spaces or inconsistent formatting in the dataset).

Boosters Carried Maximum Payload

- The names of the boosters which have carried the maximum payload mass:

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%%sql
SELECT DISTINCT "Booster_Version"
FROM SPACESTABLE
WHERE "PAYLOAD_MASS_KG_" = (
    SELECT MAX("PAYLOAD_MASS_KG_")
    FROM SPACESTABLE
);
```

```
* sqlite:///my_data1.db
Done.
```

Output

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- Explanation:**
 - ✓ The query identifies all **booster versions that carried the maximum payload mass** recorded in the dataset.
 - ✓ A **subquery** is used to first find the highest payload mass using MAX(PAYLOAD_MASS_KG_).
 - ✓ The main query then retrieves the **distinct booster versions** whose payload mass matches this maximum value.
 - ✓ DISTINCT ensures that each booster version is listed only once.

2015 Launch Records

- The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015:

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
] : %%sql
SELECT
    substr("Date", 6, 2) AS Month,
    "Booster_Version",
    "Launch_Site",
    "Landing_Outcome"
FROM SPACEXTABLE
WHERE substr("Date", 0, 5) = '2015'
AND "Landing_Outcome" = 'Failure (drone ship)';

* sqlite:///my_data1.db
Done.
```

```
] :
Month  Booster_Version  Launch_Site  Landing_Outcome
-----
01     F9 v1.1 B1012   CCAFS LC-40  Failure (drone ship)
04     F9 v1.1 B1015   CCAFS LC-40  Failure (drone ship)
```

- Explanation:**
 - ✓ The query retrieves records from **2015** where the booster experienced a **failed drone ship landing**.
 - ✓ substr(Date, 0, 5) = '2015' filters records to include only launches from the year **2015**.
 - ✓ The condition Landing_Outcome = 'Failure (drone ship)' filters the data to show only missions where the booster **attempted but failed to land on the autonomous drone ship after launch**.
 - ✓ The query displays the **month, booster version, launch site, and landing outcome**.
- Insights:**
 - In **2015**, SpaceX recorded **two failed drone ship landing attempts**, both using **Falcon 9 v1.1** boosters.
 - Both launches originated from **CCAFS LC-40**, one of SpaceX's primary launch pads at the time.

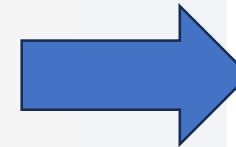
Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS OutcomeCount
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY OutcomeCount DESC;
```

```
* sqlite:///my_data1.db
Done.
```



Landing_Outcome	OutcomeCount
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Explanation:**
 - ✓ The query analyzes **landing outcomes** for SpaceX missions between **2010-06-04** and **2017-03-20**.
 - ✓ COUNT(*) calculates how many times each landing outcome occurred.
 - ✓ GROUP BY Landing_Outcome groups records by landing result.
 - ✓ ORDER BY OutcomeCount DESC ranks the outcomes from the **most frequent to the least frequent**.
 - ✓ The BETWEEN clause restricts the analysis to launches within the specified date range.
- Insight:**
 - ✓ **"No attempt"** was the most common outcome (10 occurrences), reflecting the early years when booster recovery was not routinely attempted.
 - ✓ **Drone ship landings showed equal numbers of successes and failures (5 each)**, illustrating SpaceX's rapid learning curve in developing reusable rocket technology.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

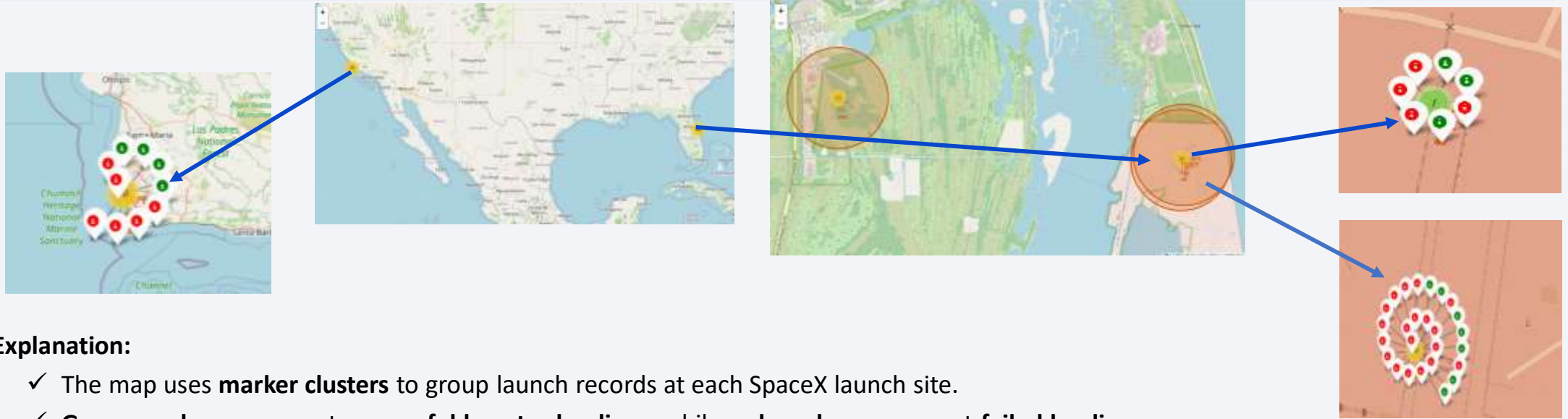
All Launch Sites on a global map

- Generated folium map and explored all launch sites' location markers on a global map:



- Are all launch sites in proximity to the Equator?
 - ✓ **No, but they are relatively closer to the Equator than many other launch locations.**
 - ✓ Florida launch sites (**CCAFS SLC-40** and **KSC LC-39A**) are located at approximately **28.5° N latitude**.
 - ✓ California's **VAFB SLC-4E** is located at approximately **34.7° N latitude**.
 - ✓ None of the sites are directly near the Equator (0° latitude), but they are positioned at lower latitudes to take advantage of the Earth's rotational speed, which helps reduce fuel consumption and improve launch efficiency.
- Are all launch sites in very close proximity to the coast?
 - ✓ **Yes.** The Florida launch sites are situated directly along the Atlantic coast.
 - ✓ VAFB SLC-4E is located on the Pacific coast.
 - ✓ The coastline-distance analysis confirms that all launch sites are only a short distance from the ocean.
- Coastal locations enhance safety**, allowing rockets to fly over open water and minimizing risk to populated areas in case of launch anomalies.
- Lower-latitude locations improve launch efficiency** by leveraging Earth's rotational velocity, reducing launch costs and increasing payload capacity.
- The clustering of launches around Florida indicates it is SpaceX's primary launch region, while California mainly supports specific mission profiles such as polar-orbit launches..

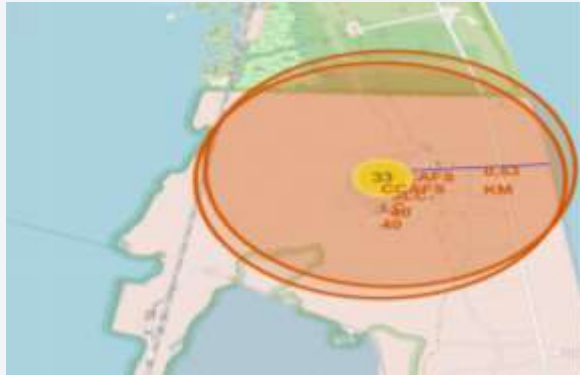
Colour-labeled Launch outcomes- Success/Failure



- **Explanation:**
 - ✓ The map uses **marker clusters** to group launch records at each SpaceX launch site.
 - ✓ **Green markers** represent **successful booster landings**, while **red markers** represent **failed landings**.
 - ✓ The numbers inside the yellow clusters indicate the total launches in that area, which expand into individual success/failure markers when zoomed in.
- **Key Findings / Insights:**
 - **CCAFS and KSC (Florida)** show a larger concentration of **green markers**, indicating **higher launch success rates** and more successful booster recoveries.
 - **VAFB SLC-4E (California)** has a mix of green and red markers, suggesting a comparatively **lower success rate** than the Florida sites.
 - The dominance of green markers across most sites highlights SpaceX's overall **high landing success performance**.
 - Launch outcome clustering reveals that **Florida-based launch sites (CCAFS and KSC)** achieved the **highest number of successful landings**, making them the strongest-performing launch locations in the dataset.

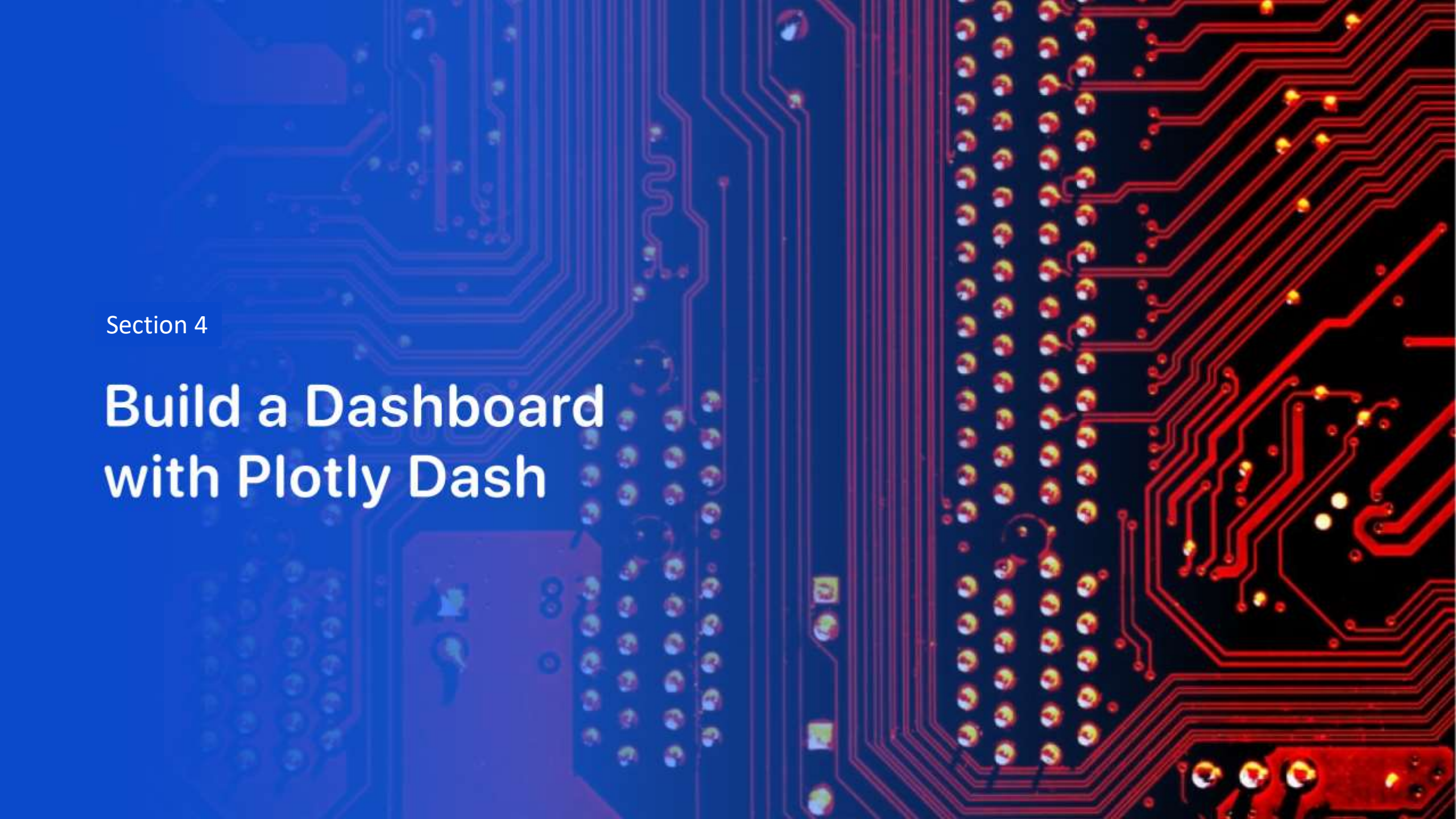
Proximities of launch sites to other points

- Launch site to its proximities such as railway, highway, coastline, with distance calculated and



- Explanation:
 - Are launch sites in close proximity to railways? **Yes**, the nearest railway line is only 1.30 kms away.
 - Are launch sites in close proximity to highways? **Yes**, the nearest highway is 0.53 kms away.
 - Are launch sites in close proximity to coastline? **Yes**, the nearest coastline is 0.69 kms away.
 - Do launch sites keep certain distance away from cities? **Yes**, launch sites are not located near major cities.
 - SpaceX launch sites maintain a safe distance from densely populated areas to minimize public safety risks and operational disruptions.





Section 4

Build a Dashboard with Plotly Dash

Launch Success Count for All Sites

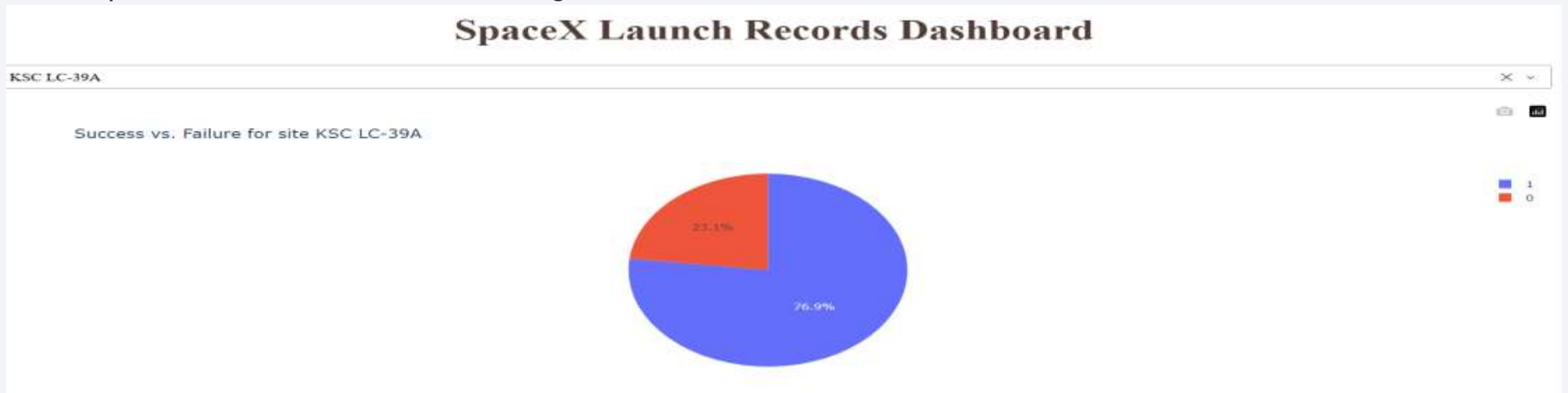
- Launch success count for all sites, in a piechart:



- **Explanation:**
 - ✓ This dashboard visualizes the **distribution of successful SpaceX launches across different launch sites**.
 - ✓ The pie chart shows the **share of total successful launches** contributed by each launch site when "All Sites" are selected.
 - ✓ Larger slices indicate a higher number of successful launches from that location.
- **Insights:**
 - ✓ **KSC LC-39A** accounts for the largest proportion of successful launches (**41.7%**), making it SpaceX's most productive launch site in this dataset.
 - ✓ **CCAFS LC-40** is the second-largest contributor with **29.2%** of successful launches.
 - ✓ **VAFB SLC-4E** and **CCAFS SLC-40** contribute smaller but significant portions of successful missions.

Launch site with Highest Launch Success Ratio

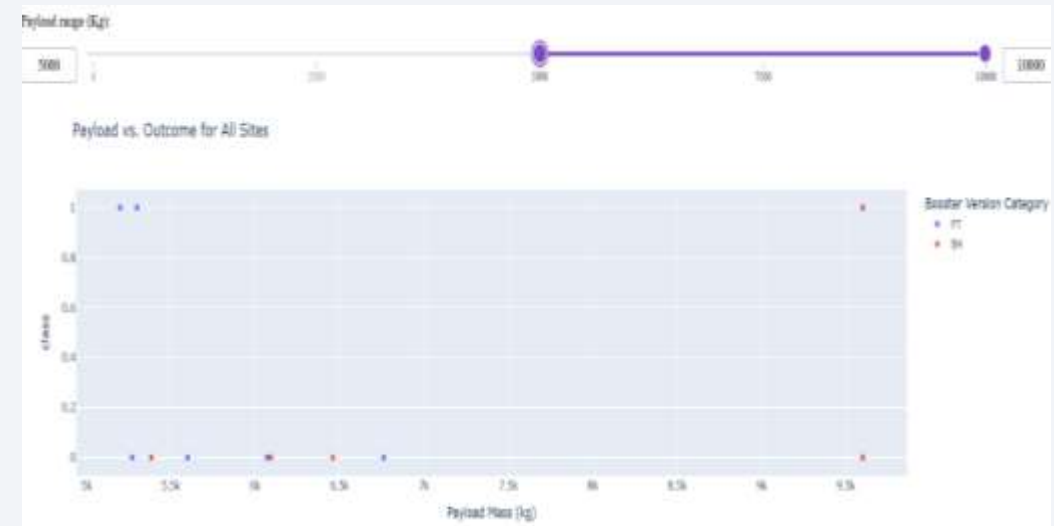
- The piechart for the launch site with highest launch success ratio:



- Explanation & Insights:**
 - ✓ This dashboard analyzes the **launch performance of KSC LC-39A**, comparing successful and failed missions.
 - ✓ **Successful launches account for 76.9%** of all missions, while **failures represent 23.1%**, indicating a strong overall success rate.
 - ✓ The large success share demonstrates the **high reliability and operational efficiency** of launches conducted from KSC LC-39A.
 - ✓ The site contributes significantly to SpaceX's launch operations and has been a key location for many successful missions.
 - ✓ Overall, the chart highlights **KSC LC-39A as one of SpaceX's most dependable launch sites**, with successful missions substantially outnumbering failures.

Payload vs Launch Outcome for All sites

- Screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider:



Explanation:

- ✓ The scatter plots show mission outcomes across different payload ranges and booster versions. Successes (1) are observed throughout both low and high payload categories.

Key Insight:

- ✓ **Payload mass has little impact on mission success**, and newer booster versions (FT/B4) maintain high success rates even for heavier payloads, demonstrating SpaceX's growing launch reliability.



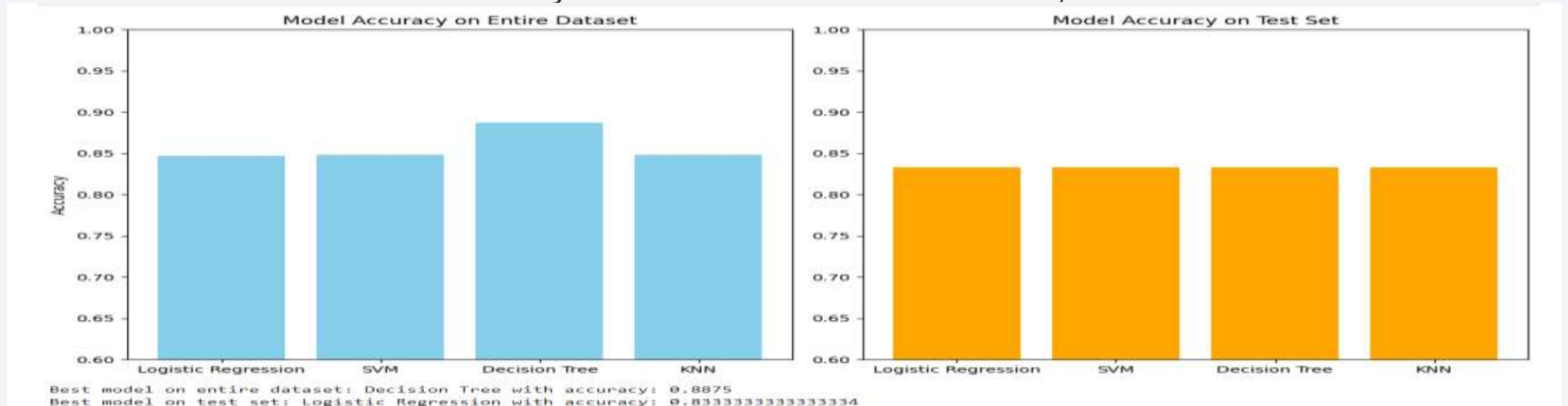


Section 5

Predictive Analysis (Classification)

Classification Accuracy

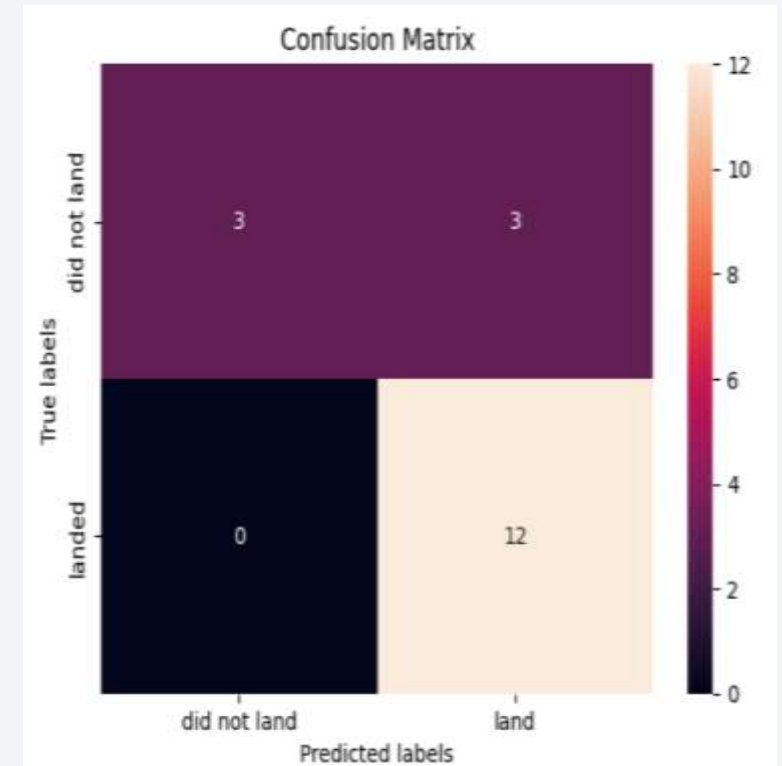
- Visualized the built model accuracy for all built classification models, in a bar chart:



- Explanation:**
 - ✓ This chart compares the **classification accuracy** of four machine learning models—**Logistic Regression, SVM, Decision Tree, and KNN**—on both the **entire dataset** and the **test dataset**.
 - ✓ **Left Chart (Entire Dataset):** Measures how well each model fits all available data.
 - ✓ **Right Chart (Test Set):** Measures how well the models perform on unseen data, which is a better indicator of real-world performance.
- Insights:**
 - ✓ **Highest accuracy on the entire dataset (training/full data):**
 - 🏆 **Decision Tree = 88.75%** (highest overall score in the chart)
 - ✓ **Highest accuracy on the test dataset (unseen data):**
 - 🏆 **Logistic Regression = 83.33%** (best model for generalization)
 - ✓ Logistic Regression has the highest classification accuracy on the test set (83.33%), making it the best model for prediction.

Confusion Matrix

- The confusion matrix of the Logistic Regression model with an explanation:
 - **Explanation:**
 - **True Negatives (TN) = 3** → Correctly predicted failed landings.
 - **False Positives (FP) = 3** → Predicted landing success, but the booster did not land.
 - **False Negatives (FN) = 0** → No successful landings were incorrectly classified as failures.
 - **True Positives (TP) = 12** → Correctly predicted successful landings.
 - **Insights:**
 - ✓ Logistic Regression correctly predicted **15 out of 18 cases (83.3% accuracy)**, with **12 successful landings and 3 failed landings classified correctly**.
 - ✓ The model achieved **100% recall for successful landings**, meaning it did not miss any actual landing successes.
 - ✓ Although it misclassified **3 failed landings as successful**, it showed the **best performance on unseen data**, indicating strong generalization and minimal overfitting.
- ❖ **Logistic Regression was the best-performing model, achieving 83.3% test accuracy and correctly identifying all successful landings, making it the most reliable model for landing outcome prediction.**



Conclusions

1. SpaceX Launches Have a High Success Rate

- Historical launch data shows that SpaceX consistently achieves successful missions, demonstrating strong operational reliability.

2. Launch Site Matters

- **KSC LC-39A** and **CCAFS LC-40** recorded the highest number of successful launches, making them key contributors to SpaceX's success.

3. Reusability Improved Over Time

- Landing outcomes evolved from failures and no-attempts to frequent successful drone ship and ground pad landings, highlighting advancements in reusable rocket technology.

4. Payload Mass Is Not a Major Limiting Factor

- Successful missions occurred across both low and high payload ranges, indicating that heavier payloads do not significantly reduce launch success.

5. Newer Booster Versions Are More Reliable

- FT and Block-series boosters showed better performance and successfully carried larger payloads, reflecting continuous technological improvements.

6. Machine Learning Can Predict Landing Outcomes

- Historical mission data contains patterns that can be used to predict booster landing success with good accuracy.

7. Logistic Regression Is the Most Reliable Model

- Logistic Regression achieved the best test accuracy (83.3%) and correctly identified all successful landings, making it the most dependable model for real-world prediction.

Appendix

- [https://github.com/Mudita13/SpaceX-ML-Prediction-Classification/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20\(1\).ipynb](https://github.com/Mudita13/SpaceX-ML-Prediction-Classification/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb)https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json
- https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_1.csv
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/labs/module_2/data/Spacex.csv
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_2.csv
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/spacex_launch_dash.csv
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/t4-Vy4iOU19i8y6E3Px_ww/spacex-dash-app.py
- https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_3.csv

Date	Time (UTC)	Booster_V	Launch_Sit	Payload	PAYLOAD_Orbit	Customer	Mission_O	Landing_Outcome
#####	18:45:00	F9 v1.0	BC	CCAFS LC- Dragon Sp	0 LEO	SpaceX	Success	Failure (parachute)
#####	15:43:00	F9 v1.0	BC	CCAFS LC- Dragon de	0 LEO (ISS)	NASA (COT)	Success	Failure (parachute)
#####	07:44:00	F9 v1.0	BC	CCAFS LC- Dragon de	525 LEO (ISS)	NASA (COT)	Success	No attempt
#####	00:35:00	F9 v1.0	BC	CCAFS LC- SpaceX CR	500 LEO (ISS)	NASA (CRS)	Success	No attempt
#####	15:10:00	F9 v1.0	BC	CCAFS LC- SpaceX CR	677 LEO (ISS)	NASA (CRS)	Success	No attempt
#####	16:00:00	F9 v1.1	B1	VAFB SLC- CASSIOPE	500 Polar LEO	MDA	Success	Uncontrolled (ocean)
#####	22:41:00	F9 v1.1	CCAFS LC-	SES-8	3170 GTO	SES	Success	No attempt
#####	22:06:00	F9 v1.1	CCAFS LC-	Thaicom 6	3325 GTO	Thaicom	Success	No attempt
#####	19:25:00	F9 v1.1	CCAFS LC-	SpaceX CR	2296 LEO (ISS)	NASA (CRS)	Success	Controlled (ocean)
#####	15:15:00	F9 v1.1	CCAFS LC-	OG2 Missi	1316 LEO	Orbcomm	Success	Controlled (ocean)

SpaceX Launch Records Dashboard									
All Sim									
Q Search									
All Sim									
CCAFS LC-49									
VAFB SLC-4E									
BSC LC-39A									
CCAFS SLC-4B									

Thank you!

